Matrix functions and stability

Uniqueness of positive definite solutions of two-letter word equations in order two

Difficulty: challenging **Importance:** interesting to specialist

Status: Partially resolved

Topic: Structured nonlinear matrix equations.

Rating rationale: Challenging because uniqueness must hold for words of unbounded length despite higher-dimensional failures; specialist impact reflects the surviving two-letter, order-two setting.

Problem statement

A word $W(X, B)$ is a finite product of letters from $\{X, B\}$. It is symmetric if its sequence of letters equals its reversal. Require at least one occurrence of X.

For every such symmetric word W and every pair of Hermitian positive definite matrices $B, P \in \mathbb{C}^{2\times 2}$, is there exactly one Hermitian positive definite matrix $X \in \mathbb{C}^{2\times 2}$ satisfying

$$W(X, B) = P?$$

Products are evaluated in the written order. Existence is known; uniqueness is the open assertion. This is the ordinary two-letter word case of the surviving order-two conjecture, with no real powers or additional fixed letters included in the statement.

Why it matters

The question concerns whether a structured nonlinear matrix equation has a single positive definite solution branch. The elementary word XBX connects this family with matrix geometric means and Riccati equations.

References

- C. J. Hillar and C. R. Johnson, Symmetric word equations in two positive definite letters, *Proceedings of the American Mathematical Society* 132 (2004), 945–953, Definition 1.1, Theorem 2.2, and Conjecture 2.3. This gives the original two-letter existence and uniqueness formulation.
- S. N. Armstrong and C. J. Hillar, Solvability of Symmetric Word Equations in Positive Definite Letters, *Journal of the London Mathematical Society* 76 (2007), 777–796, Theorem 1.4 and Conjecture 11.5 (p. 20 of the arXiv PDF); Remark 11.6 discusses the real order-two case.
- J. D. Lawson and Y. Lim, Solving symmetric matrix word equations via symmetric space machinery, *Linear Algebra and its Applications* 414 (2006), 560–569. Its bounded-degree uniqueness result is summarized in Armstrong–Hillar's introduction.

Status check

On 2026-09-08, searched the titles, "symmetric word equations", "Conjecture 11.5", "2×2", "two-by-two", "uniqueness", "counterexample", and 2025–2026. No order-two resolution was located. Armstrong–Hillar refute unrestricted uniqueness in dimensions at least three, while explicitly retaining the order-two conjecture; their special order-two theorem and known bounded-degree results do not cover every word. No recent explicit reaffirmation was found. The statement deliberately records the source-backed two-letter problem rather than presuming that the paper's comments about reducing complex matrices to real matrices cover arbitrarily many fixed letters.

Audit — 2026-09-10

Rechecked Armstrong–Hillar, Theorem 11.4 and Conjecture 11.5. The nontrivial word $XBX^2B^3X^2BX$ is settled in order two, but arbitrary words remain open. Title and order-two uniqueness searches found no full resolution. Existence alone is not the reason for the partial-resolution tag.